Logistic Regression – Airline Customer Satisfaction Prediction

Fully executed notebook with preprocessing, encoding, logistic regression, evaluation metrics, coefficient interpretation, recommendations, and limitations.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, classification_report

df = pd.read_csv('invistico_Airline.csv')
print("Shape:", df.shape)
df.head()
```

```
Shape: (129880, 22)
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	Inflight wifi service	Inflight entertainment
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	0	0	2	2	4
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	0	0	3	0	2
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	0	0	3	2	0
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	0	0	3	3	4
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	0	0	3	4	3

```
print(df['satisfaction'].value_counts())
```

```
satisfaction
satisfied       71087
dissatisfied    58793
Name: count, dtype: int64
```

```
X = df.drop('satisfaction', axis=1)
y = (df['satisfaction'].astype(str).str.lower()=='satisfied').astype(int)

cat_cols = X.select_dtypes(include='object').columns.tolist()
num_cols = [c for c in X.columns if c not in cat_cols]

preprocessor = ColumnTransformer([
    ('num', Pipeline([
        ('imputer', SimpleImputer(strategy='median')),
        ('scaler', StandardScaler())
    ]), num_cols),
    ('cat', Pipeline([
        ('imputer', SimpleImputer(strategy='most_frequent')),
        ('onehot', OneHotEncoder(handle_unknown='ignore'))
    ]), cat_cols)
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(max_iter=1000))
])

model.fit(X_train, y_train)
pred = model.predict(X_test)

cm = confusion_matrix(y_test, pred)
print("Confusion Matrix\n", cm)
print("Accuracy:", accuracy_score(y_test,pred))
print("Precision:", precision_score(y_test,pred))
print("Recall:", recall_score(y_test,pred))
print(classification_report(y_test,pred))
```

```
Confusion Matrix
 [[ 9544  2215]
 [ 2233 11984]]
Accuracy: 0.8287650138589467
Precision: 0.8440030988097753
Recall: 0.8429345150172329
              precision    recall  f1-score   support

           0       0.81      0.81      0.81     11759
           1       0.84      0.84      0.84     14217

    accuracy                           0.83     25976
   macro avg       0.83      0.83      0.83     25976
weighted avg       0.83      0.83      0.83     25976
```

```
feature_names = model.named_steps['preprocessor'].get_feature_names_out()
coef = model.named_steps['classifier'].coef_[0]

coef_df = pd.DataFrame({'Feature':feature_names,'Coefficient':coef})
coef_df['Abs'] = coef_df['Coefficient'].abs()

print("Top Positive Drivers")
print(coef_df.sort_values('Coefficient', ascending=False)[['Feature','Coefficient']].head(10))

print("\nTop Negative Drivers")
print(coef_df.sort_values('Coefficient', ascending=True)[['Feature','Coefficient']].head(10))
```

```
Top Positive Drivers
                                Feature  Coefficient
7           num__Inflight entertainment     0.972683
18    cat__Customer Type_Loyal Customer     0.826756
22                  cat__Class_Business     0.423151
2                     num__Seat comfort     0.392571
10                num__On-board service     0.388175
13                 num__Checkin service     0.354962
9           num__Ease of Online booking     0.333694
11                num__Leg room service     0.308907
20  cat__Type of Travel_Business travel     0.267379
15                 num__Online boarding     0.181688

Top Negative Drivers
                                   Feature  Coefficient
19    cat__Customer Type_disloyal Customer    -1.046129
21     cat__Type of Travel_Personal Travel    -0.486751
24                      cat__Class_Eco Plus    -0.349142
3   num__Departure/Arrival time convenient    -0.327559
4                      num__Food and drink    -0.296607
23                           cat__Class_Eco    -0.293382
17           num__Arrival Delay in Minutes    -0.258479
1                      num__Flight Distance    -0.181496
0                                 num__Age    -0.133204
6               num__Inflight wifi service    -0.127863
```

Business Recommendations

- Improve inflight entertainment and seat comfort.
- Strengthen loyalty programs.

- Improve digital booking and check-in experience.
- Focus on service quality and economy-class experience.

Limitations

- Logistic Regression assumes linearity in log-odds.
- Additional models such as Random Forest and XGBoost should be evaluated.